

Branch-wise versus clade-grouped gene-family evolution on the archaeal tree

Reconciliation of 7,059 gene families on the 257-taxon archaeal species tree, Euryarchaeota root

Reconciling each gene family against the species tree under a duplication–transfer–loss (DTL) model requires a choice of how finely rates may vary across the tree. We compared two parameterisations: the *clade-grouped* model, in which all branches of a major clade share a single set of duplication, loss and transfer rates (and origination weight), and a *fully branch-wise* model, in which every branch carries its own rates. As a control, our GPU reconciliation engine reproduces the reference per-family likelihoods to within Pearson $r = 0.999998$, so the comparison reflects the rate model, not the implementation.

Relaxing the clade constraint improves the fit by $\sim 10,900$ log-likelihood units (Table 1) while leaving the inferred root unchanged: Euryarchaeota remains the maximum-likelihood rooting under both models. The improvement is not overfitting — on gene families withheld from rate estimation, the branch-wise model still predicts better than the clade-grouped model (+1,556 units; +0.44 per family), so the additional parameters capture real among-branch rate variation.

That variation has a clear and interpretable structure (Figure 1). Duplication and transfer rates are essentially *clade-constant*: within each named clade the per-branch estimates scarcely depart from a single value (clade membership explains $R^2 \approx 0.94$ of their variance). Loss is markedly less clade-structured ($R^2 \approx 0.38$): its extra variation is distributed across branches rather than captured by clade identity. Biologically, gene gain (duplication, transfer) is well summarised by one rate per clade, whereas gene loss is more branch-idiosyncratic.

The reconstructed ancestral genome is robust to this added flexibility (Figure 2). Of the 7,059 families, the branch-wise model assigns 903 to the last archaeal common ancestor and the clade-grouped model 772; 722 are common to both (Jaccard 0.76), and the per-family probabilities of origination at the root agree closely ($r = 0.96$). The two models therefore recover essentially the same ancestral gene set, the branch-wise model reconstructing a modestly larger genome.

rate model	free params	data log L (nats)	AU 0.05 root set
clade-grouped (one D/L/T per clade)	~ 70	$-1,642,425$	{Euryarchaeota, MHH} (region)
fully branch-wise (per branch)	$\sim 2,000$	$-1,631,537$	{Euryarchaeota}
branch-wise – grouped: +10,888 in-sample; +1,556 on held-out families (cross-validated, not overfitting).			

Table 1: Model comparison at the Euryarchaeota root. Both models recover Euryarchaeota as the maximum-likelihood root; the approximately-unbiased (AU) test supports a {Euryarchaeota, MHH} region under the parsimonious clade-grouped model and narrows to Euryarchaeota alone under the richer branch-wise model — an over-confidence we attribute to model richness, not to stronger evidence.

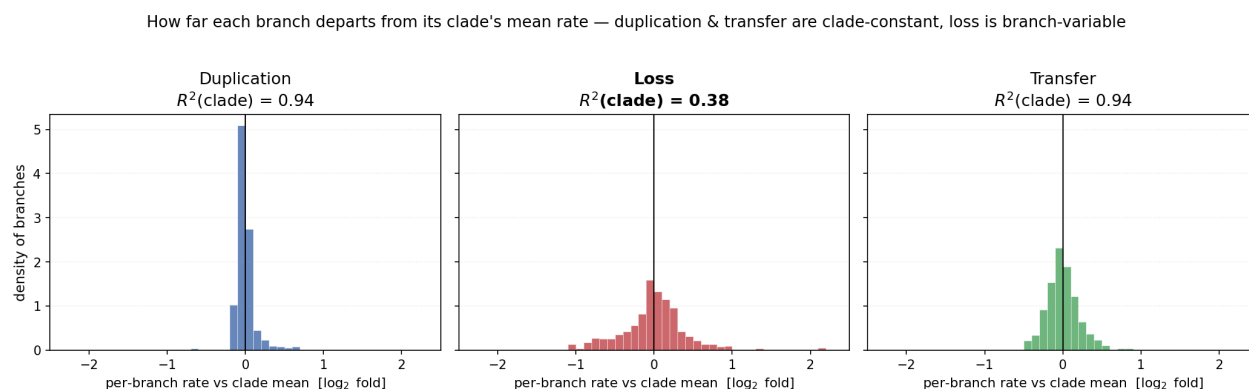


Figure 1: Departure of each branch's fitted rate from its clade's mean rate ($\log_2$ fold; shared axes). Duplication and transfer form a sharp spike at zero — the per-branch estimates scarcely leave their clade mean, so clade membership explains $R^2 \approx 0.94$ of the variance — whereas loss is broad ($R^2 \approx 0.38$), i.e. branch-specific. Gene gain (duplication, transfer) is well summarised by one rate per clade; gene loss is not.

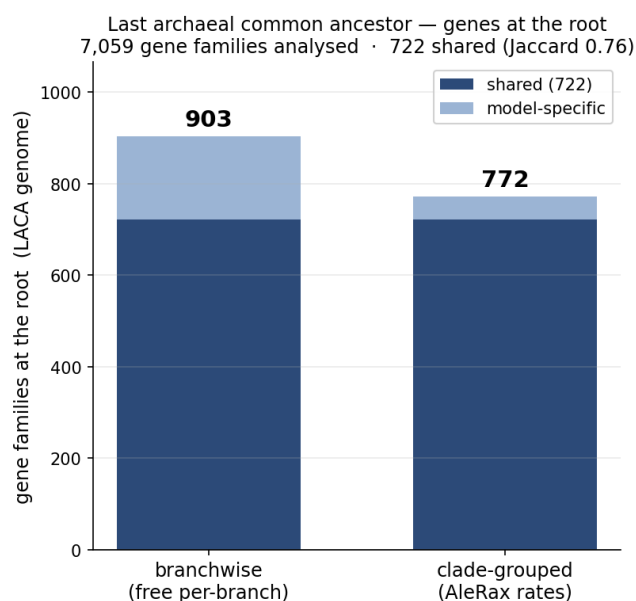


Figure 2: Reconstructed last archaeal common ancestor (LACA) gene content: 903 families under the branch-wise model vs. 772 under clade-grouped rates, 722 shared (Jaccard 0.76), of 7,059 analysed.